



Effects of nanotubes pore size on the catalytic performances of iron catalysts supported on carbon nanotubes for Fischer–Tropsch synthesis

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ABSTRACT

In this report, the effects of pore diameter and structure of iron catalysts supported on carbon nanotubes (CNTs) on the reaction rates and product selectivities of Fischer–Tropsch (FT) reactions are presented. In order to study the effects of pore diameter, two types of CNTs with different average pore sizes (12 and 63 nm), however, with similar surface areas were prepared. The iron catalysts were prepared using incipient wetness impregnation method and characterized by ICP, BET, XRD, TPR, SEM and TEM analyses. According to the TEM images of iron catalysts supported on narrow pore CNTs (Fe/np-CNT) and wide pore CNTs (Fe/wp-CNT), a vast majority (~80%) of the iron oxide particles were deposited inside the nanotubes' pores. The TEM and XRD analysis showed that the iron oxide particles on the Fe/wp-CNT (17 nm) were larger than those on Fe/np-CNT sample (11 nm). Temperature programmed reduction analyses of the catalysts showed that the extent of reduction of the Fe/np-CNT catalyst was 17% higher compared to that of the Fe/wp-CNT catalyst. Finally, catalytic performances of both catalysts were evaluated in a fixed-bed reactor for FT reactions at 2 MPa and 275 °C. At these conditions, the activity of the np-CNT catalyst (%CO conversion of 30) was 2.5 times that of the wp-CNT catalyst (%CO conversion of 12). In addition, the Fe/wp-CNT was more selective toward lighter hydrocarbons with a methane selectivity of 41% compared to that of the np-CNT catalyst with methane selectivity of 14.5%. Deposition of metal particle on the CNT with narrow pore size resulted in more active and selective catalyst due to higher degree of reduction and higher metal dispersion.

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1. Introduction

Fischer–Tropsch (FT) synthesis is a potentially attractive technology for production of clean liquid fuels from syngas. The syngas can be supplied through gasification of coal or biomass and reforming of natural gas. FT synthesis proceeds on metal (Fe, Co and Ru) supported catalysts. The efficiency of FT synthesis can be enhanced by design of new catalysts with higher syngas conversion, higher C₅+ yield, and lower methane selectivity [1,2]. In FT processes, the catalyst activity and selectivity are influenced by nature and structure of support, nature of metal, metal dispersion, metal loading, and catalyst preparation method [3]. A large number of studies on

FT catalysts have been carried out using active metals supported on silica, alumina or titania [4]. In recent years, carbon nanotubes (CNTs) have been considered as possible new support for FT reactions [5–11].

Carbon nanotubes with unique properties such as uniform pore size distribution, meso and macro pore structure, inert surface properties, and resistance to acid and base environment can play an important role in a large number of catalytic reactions [11]. CNTs are essentially composed of graphite layers with a tubular morphology [7,8]. Structural parameters of CNT such as inner and outer diameter and length of nanotubes can be manipulated using different synthesis processes and operating conditions. It has been shown that CNT supports offer improved performance for FT reactions [4–9,12]. In application of CNT supports, the effects of pore size and diameter of carbon nanotubes on the catalytic performance of FT catalysts have not been elucidated yet. In general, pore size of supported catalysts can influence particle size distribution, dispersion, extent of reduction, and mass transfer rates [4].

In this report, the effects of pore diameter of CNTs on FT reaction rates and selectivities over iron catalysts are presented. Two types of CNTs with different average pore sizes (12 and 63 nm) with comparable surface areas were used.

Abbreviations: AAO, anodic aluminum oxide; BET, Brunauer, Emmett, and Teller; CNT, carbon nanotubes; CVD, chemical vapor deposition; FTS, Fischer–Tropsch synthesis; ICP, inductively coupled plasma; SEM, scanning electron microscopy; SL, standard litter; TEM, transition electron microscopy; TOS, time on stream; TPR, temperature programmed reduction; WGS, water–gas shift; XRD, X-ray diffraction; Fe/wp-CNT, iron catalysts deposited on wide carbon nanotubes; Fe/np-CNT, iron catalysts deposited on narrow pore carbon nanotubes.

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Nomenclature

P	pressure (MPa)
T	temperature (°C)
%CO	percent of CO conversion (mol%)

2. Experimental

In order to study the effects of inner pore size of CNTs on the performance of iron catalyst for the FT reactions, two CNT samples with considerable different pore diameters were used. The narrow pore CNT sample (denoted as np-CNT) was purchased from mknano Co. (M.K. IMPEX Canada). The wide pore CNT sample (denoted as wp-CNT) were synthesized in our laboratories using anodic aluminum oxide (AAO) films along with chemical vapor deposition method [12]. The procedure for production of wp-CNT is as follows:

The AAO templates were synthesized using the two-step anodization procedure. The electrolysis cell consisted of a graphite cathode and a pure (99.5%) aluminium electrode. A DC power was applied to the cell and the electric potential across the electrodes was kept constant for the duration of experimental run. The electrolysis vessel contains 1.3 L of oxalic acid electrolyte solution at a concentration of 0.40 M. The first anodization was then performed with the polished anode at a constant potential of 40.0 V for 1 h. The anode was then removed and immersed for 40 min in a deoxidation solution at 70 °C. The deoxidation solution consisted of the following mass composition: 92.1% H₂O, 7.0% H₃PO₄, and 0.9% CrO₃.

For the second anodization, the energy potential was kept constant at 40.0 V for a run time of 66 h. After the second anodization was completed, the electrode terminals were exchanged and a constant current of 2.00 A was applied through the cell to extract the AAO product from the anode surface. To partially remove alumina from the AAO (for optimum pore diameter), each batch of template was immersed in a 5% H₃PO₄ solution for 30 min.

The wp-CNT sample was synthesized from the AAO template in a chemical vapor deposition (CVD) process. The reactor was heated to a constant temperature of 650 °C across the length of the template boats under a constant Ar gas flow of 100 mL/min. Once the reaction temperature was reached, acetylene gas was introduced to the system over a span of 2 h. During this period, Ar gas flow was maintained constant at ratio of 2.5 mL Ar/min: 1.0 mL C₂H₂/min. After 2 h, the acetylene flow was stopped while argon flow at 100 mL/min was continued as the reactor was cooled over a 6-h period. To remove the synthesized CNTs from the pre-existing AAO template,

the combined product was magnetically stirred in 250 mL of 23% HF solution for 30 h. Then, the insoluble carbon was filtered from the solution and dried, leaving a pure CNT product.

Both wp-CNT and np-CNT samples were treated in nitric acid (60 wt%) at 110 °C for 16 h. The catalysts were prepared according to the incipient wetness impregnation method. For the preparation of the catalyst with iron particles inside the nanotube pores with 20 wt% Fe, a specified amount of Fe(NO₃)₃·9H₂O was dissolved in deionised water. The amount of solution was equal to total pore volume of the CNT samples. In this case, the metal solution filled the inner pores of CNTs due to capillary effects [5]. After drying at 120 °C followed by calcination at 400 °C for 3 h, the catalysts were characterized by nitrogen adsorption, inductively coupled plasma (ICP), temperature programmed reduction (TPR), X-ray diffraction (XRD), scanning electron microscopy (SEM) and transmission electron microscopy (TEM).

The surface area and pore volume of the CNT supports and catalysts were measured by an ASAP-2000 Micromeritics system. Before analysis, the samples were degassed at 200 °C for 2 h under 50 mTorr vacuum. X-ray diffraction (XRD) analyses were performed using a Philips PW1840 X-ray diffractometer with monochromatized CuK α radiation. The crystallite diameter was determined by substituting the half-width of a chosen peak into the Debye–Scherrer equation.

The reduction behavior of the catalyst precursors was studied by temperature programmed reduction (TPR) using a CHEMBET-3000 equipped with a thermal conductivity detector. 0.1 g of catalyst was placed in U-shaped quartz tube. A 5% hydrogen/nitrogen mixture was introduced (flow rate = 36 cm³ (STP)/min) and the furnace temperature was ramped from room temperature to 1173 K at 10 K/min.

The morphology of samples was studied by transmission electron microscopy (TEM). Sample specimens for TEM studies were prepared by ultrasonic dispersion of the catalysts in ethanol and the suspensions were dropped onto a copper grid. TEM micrographs were obtained using a Hitachi H-7500 (120 kV). Several TEM micrographs were recorded for each sample and analyzed to determine the particle size distribution. A Phillips SEM-505 scanning electron microscope operating at 300 kV in SE display mode was also used. For characterization prior to analysis, all the samples were gold coated in a sputter coating unit (Edwards Vacuum Components Ltd., Sussex, England).

The Fischer–Tropsch synthesis was performed in a fixed-bed micro reactor. Prior to CO hydrogenation, in-situ reduction was conducted according to the following procedure. The catalyst (0.5 g) was placed in the reactor and diluted with 5 g silicon carbide. Then

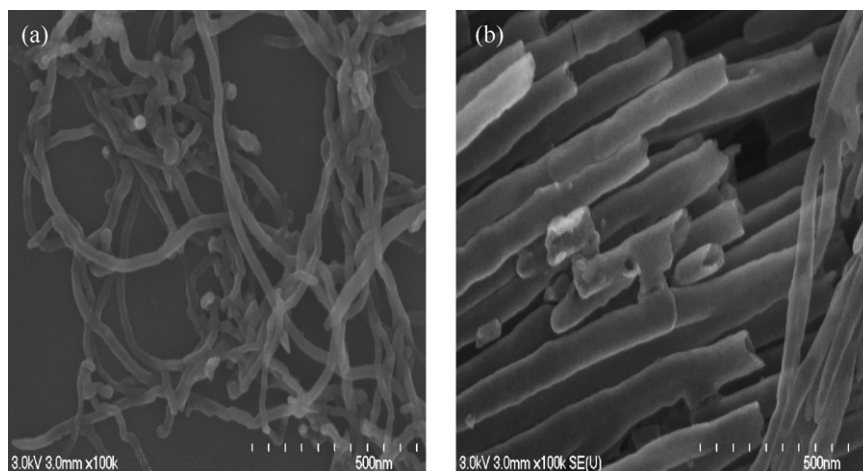


Fig. 1. SEM micrograph of the CNT supports, (a) np-CNT, (b) wp-CNT.

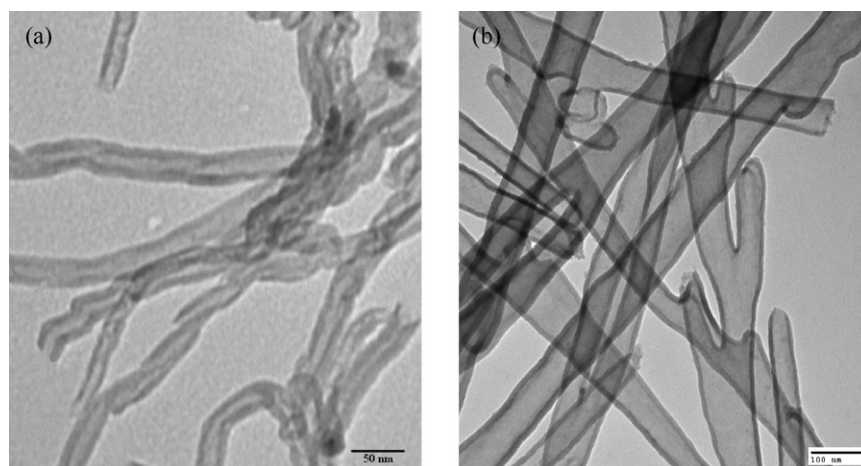


Fig. 2. TEM images of CNT supports (a) np-CNT, (b) wp-CNT.

pure hydrogen was introduced into the reactor at a flow rate of 45 mL/min. The reactor temperature was increased from room temperature to 380 °C (at a rate of 1 °C/min) and maintained at this temperature for 14 h. Then, the reactor temperature was decreased to 275 °C under flowing hydrogen. All gas flow rates were controlled by Brooks 5850 mass flow controllers. Argon was used as internal standard gas in the reactor feed. The mixed gases (30%CO, 60%H₂, 10% Ar) entered through the top of the fixed-bed reactor. The temperature of the reactor was controlled via a PID temperature controller. Synthesis gas with a space velocity of 7200 cm³ (STP)/(h g) was introduced and the reactor pressure was increased to 2 MPa. Products were continuously removed from the vapor and passed through two traps. The pressure of uncondensed vapor stream was reduced to atmospheric pressure. The composition of the outlet gas stream was quantified using an on-line GC-2014 Shimadzu gas chromatograph. The contents of traps were removed every 24 h and the hydrocarbon and water fractions were separated, and analyzed by Varian 3400 GC. Catalytic activity and product selectivity were calculated after a time on stream of 50 h.

3. Results and discussions

3.1. Characterization of supports and catalysts

Scanning electron micro graphs (Fig. 1a and b) of the acid-treated and purified CNTs samples (np-CNT and wp-CNT) revealed

that the samples contained CNTs and no other impurities were observed. Fig. 1a shows that the np-CNT sample contained woven nanotubes with high aspect ratios (ratio of length to outer diameter of nanotubes is more than 25). For wp-CNT sample, SEM image (Fig. 1b) indicated that nanotubes were well-aligned and the caps of nanotubes were open.

Transmission electron micro graphs (TEM) of the acid-treated CNT samples are shown in Fig. 2a and b. Fig. 2a shows that the np-CNT nanotubes sample had uniform diameter and their inner and outer diameters varied between 8–12 and 20–25 nm, respectively. TEM analysis also revealed that a vast majority (more than 70%) of the np-CNT sample had open caps (Fig. 2a). As seen in Fig. 2b, the wp-CNT sample comprised of straight nanotubes with inner diameter of 50–70 nm along with several Y junction with inner diameter of 20–50 nm. The thickness of the walls varied between 7 and 9 nm.

The particle size distribution and position of iron particles inside or outside of the CNTs were studied using several TEM images from each of the catalyst samples. Fig. 3a and b show representative TEM images of iron catalysts supported on narrow pore CNT sample (Fe/np-CNT) and iron catalyst supported on wide pore CNT sample (Fe/wp-CNT). Dark spots represent the iron oxide particles which are attached inside or outside of the nanotubes. For both catalysts, a vast majority of iron particles (80%) were distributed in the inner pores of CNTs (Fig. 3a and b). This can be attributed to carbon nanotubes' tubular structure, which can induce capillary forces during the impregnation process. In the case of Fe/np-CNT iron oxide par-

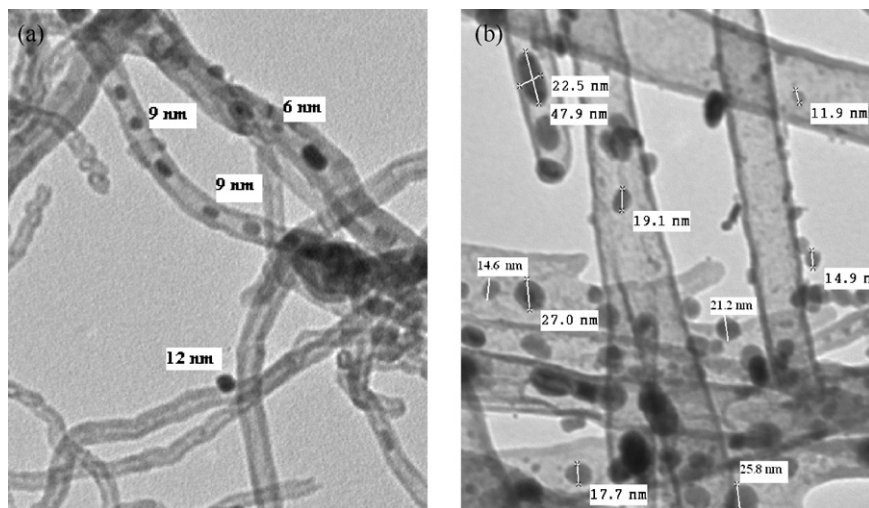


Fig. 3. TEM images of the iron catalysts (a) Fe/np-CNT (b) Fe/wp-CNT. Dark spots represent the iron oxide particles inside and out side of the nanotubes.

Table 1

Metal content and surface properties of CNT samples, Fe/np-CNT and Fe/wp-CNT catalysts.

Catalyst name	Metal content% ^a	BET surface area (m ² /g)	Total pore volume (cm ³ /g)
np-CNT (support)	0	212	0.58
wp-CNT (support)	0	218	0.55
Fe/np-CNT	19.6	155	0.44
Fe/wp-CNT	20.1	186	0.48

^a Determined by ICP analysis.

ticles varied in 4–14 nm range with most abundant particles with a diameter of 8–10 nm. In the case of Fe/wp-CNT catalyst, particle sizes varied from 5 to 40 nm with most particles were in the range of 14–25 nm.

The data suggest that inner pore of nanotubes physically restricted the particle size growth located inside these nanotubes.

Table 1 shows the metal content for the Fe/np-CNT and Fe/wp-CNT catalysts. Inductively coupled plasma (ICP) analyses of the catalysts revealed that the metal contents of the catalysts were close ($\pm 0.4\%$) to the target metal content of 20 wt% Fe. Table 1 also shows the results of surface area measurements of both CNT supports and corresponding iron catalysts. As discussed earlier, both CNT samples had a comparable surface areas (212 m²/g for np-CNT and 218 m²/g for wp-CNT). As a result, both Fe/np-CNT and Fe/wp-CNT catalysts held similar metal loading per surface area of the support. According to the N₂ adsorption analysis, the loading of 20% Fe decreased the surface area of Fe/np-CNT and Fe/wp-CNT catalysts to 155 and 186 m²/g, respectively. These data show that the BET surface area of the catalysts were lower than that of the corresponding supports indicating pore blockage due to iron loading on the supports. However, comparison between BET surface area of the Fe/np-CNT and Fe/wp-CNT catalysts showed that the extent of pore blockage in the Fe/np-CNT was higher than that in Fe/wp-CNT.

Fig. 4 shows XRD patterns of the calcined Fe/np-CNT and Fe/wp-CNT catalysts. For both catalysts, the peaks at 2θ values of 26° and 44° correspond to the graphite layers of multi-walled nanotubes. Surprisingly, the Fe/np-CNT and Fe/wp-CNT catalyst did not show similar XRD patterns. According to the XRD spectra, the diffraction peaks for Fe/np-CNT catalyst matched very well with the standard Hematite (Fe₂O₃) phase, whereas, for Fe/wp-CNT catalysts, the peaks at 2θ of 30 and 35.7° indicated the presence of standard Magnetite (Fe₃O₄) phase. This can be attributed to the different electronic properties of inner surface of CNTs with different pore diameters [13,15].

Table 2 shows the average iron oxide particle sizes on the catalysts calculated from XRD spectrum using Debye–Scherrer equation at the most intense peak of 36°. The data verified that the iron oxide particles on Fe/wp-CNT (19 nm) are larger than that on the Fe/np-

Table 2

Particle sizes (iron oxide) based on XRD and TEM analysis.

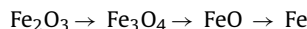
Catalyst	Particle size-XRD (nm)	Particle size-TEM (nm)
Fe/np-CNT	11 $\pm$ 2 ^a	8–10 ^a
Fe/wp-CNT	19 $\pm$ 1 ^b	14–25 ^b

^a Fe₂O₃.

^b Fe₃O₄.

CNT catalysts (11 nm). To compare the particle size calculated based on XRD with TEM analysis, the size distribution of the particles is also shown in Table 2. There is a good agreement between the data for average particle size calculated from XRD and the most abundant sizes from TEM analysis.

TPR analyses were performed to evaluate the reducibility of the Fe/np-CNT and Fe/wp-CNT catalysts. TPR patterns of Fe/np-CNT and Fe/wp-CNT catalysts are shown in Fig. 5. Four peaks (A, B, C and D) can be observed on the TPR profile of the Fe/np-CNT catalyst, whereas, two peaks appear on the TPR profile of the Fe/wp-CNT catalysts. Generally, the reduction of iron oxides takes place according to the following scheme [10]:



In the case of Fe/np-CNT, the first peak (A) can be assigned to the reduction of Fe₂O₃ to Fe₃O₄. The second peak, B, can be assigned to the subsequent reduction of Fe₃O₄ to FeO. Peak C, observed at 500–700 °C, can be related to the reduction of FeO to metallic Fe. Peak D can be attributed to gasification of CNTs at a temperature higher than 600 °C. For Fe/wp-CNT catalyst, since the iron oxide was in the form of Fe₃O₄, the first peak (B) can be related to the reduction of Fe₃O₄ to FeO and the second peak, C, can be assigned to the subsequent reduction of FeO to Fe. There is a tail (after 650 °C) for gasification of wp-CNT as seen in the TPR profile of the Fe/wp-CNT catalyst.

Table 3 shows reduction temperatures and extent of reduction for both catalysts. According to the reduction temperature, deposition of iron oxide particles inside the nanotubes with narrow pores results in a decrease in the temperature of the first TPR peak from 459 to 438 °C. However, the onset of second peak (C) was delayed from 574 to 667 °C compared to Fe/wp-CNT. Delay in second peak has been reported for different supported iron catalysts [16]. The extent of reduction of the metal is the ratio of the actual H₂ consumed during the temperature programmed reduction (from ambient temperature to 700 °C) process to the calculated theoretical amount of H₂ needed for the complete reduction of the metal oxides.

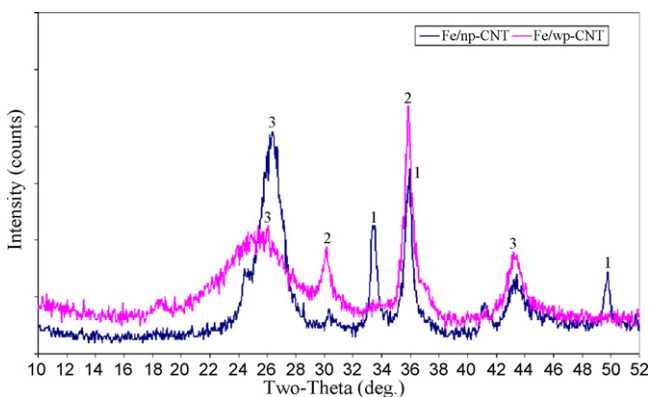


Fig. 4. XRD spectra of Fe/np-CNT and Fe/wp-CNT (1: Fe₂O₃, 2: Fe₃O₄, 3: CNT).

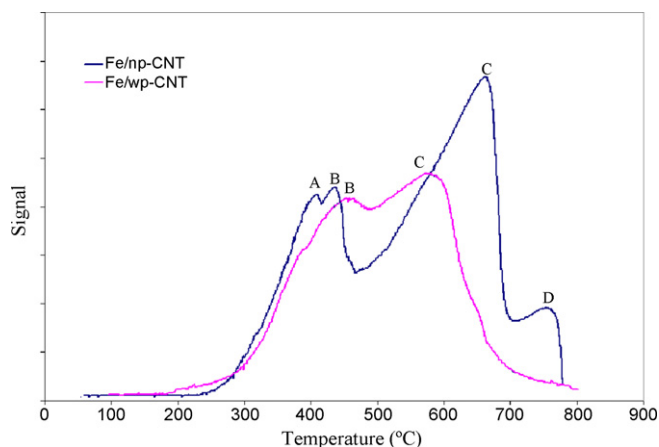


Fig. 5. H₂-TPR profiles for Fe/np-CNT and Fe/wp-CNT catalysts.

Table 3
Reduction temperature and extent of reduction (from 25 to 700 °C) based on H₂-TPR analysis.

Catalyst	Peak A	Peak B (°C)	Peak C (°C)	Peak D (°C)	Extent of reduction (25–700 °C) ^a
Fe/np-CNT	401 ± 5	438 ± 3	667 ± 6	747 ± 8	79 ± 4
Fe/wp-CNT	–	459 ± 4	574 ± 5	–	62 ± 3

^a Values for extent of reduction were calculated by considering the fact that the iron oxides on Fe/np-CNT and Fe/wp-CNT were in the form of Fe₂O₃ and Fe₃O₄, respectively.

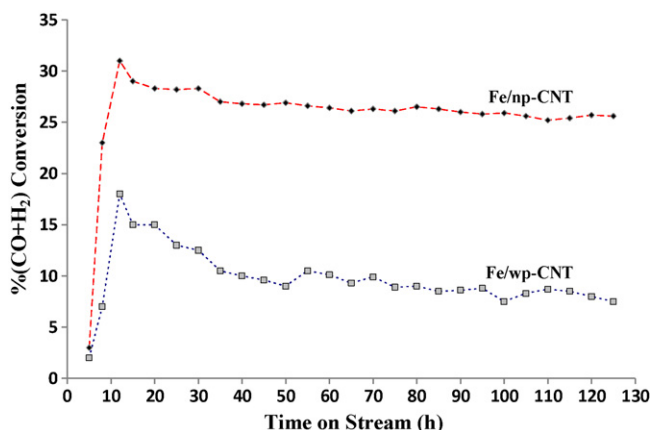


Fig. 6. Variation of catalytic activity for Fe/np-CNT and Fe/wp-CNT catalysts with time on stream.

According to Table 3, the extent of reduction of Fe/np-CNT (79%) was higher than that of Fe/wp-CNT (62%). This can result in relatively more incomplete reduction of the metal sites in the case of the Fe/wp-CNT catalyst and lower FT activity. According to the literature, in the case of FT catalysts with large particles, the particle size does not affect the extent of reduction [4]. Theoretical studies [13] and our previous work [14] indicate that deviation of the graphene layers from planarity causes π -electron density to shift from the concave inner surface to the convex outer surface leading to different electronic properties of substances in contact with either inner and outer surfaces of CNTs. Considering the different temperature programmed reduction profiles for iron oxides doped on inner pore of CNTs with different pore diameter, it can be postulated that the extent of shift for electron density could be more for a nanotube with narrow pore compared to a wide nanotube. Thus, this phenomenon may be attributed to a different interaction of iron oxide with interior surface of the Fe/np-CNT catalyst resulting in higher extent of reduction of this catalyst.

Table 4

CO conversion and product selectivity of Fe/np-CNT and Fe/wp-CNT.

Catalyst activity and product selectivity	Fe/np-CNT	Fe/wp-CNT
(CO + H ₂) %	25 ± 2	10 ± 1
CH ₄	14.5 ± 0.5	41.0 ± 0.5
C ₂ –C ₄	37.0 ± 0.5	47.0 ± 1
C ₅ +	48.5 ± 1	12.0 ± 1
CO ₂	19.5 ± 2	15.0 ± 1

Process conditions: 7.2 Sl/g-cat/h, 2MPa, H₂/CO = 2, TOS = 120 h, all catalysts contain 20 wt% Fe, HC in wt%; CO₂ in mol%.

4. Fischer–Tropsch synthesis

The performances of the Fe/np-CNT and Fe/wp-CNT catalysts for the FT synthesis were evaluated in a fixed-bed reactor. All the reactions were performed under a set of similar conditions (275 °C, 2 MPa, H₂:CO = 2). CO hydrogenation (blank runs with no iron) was performed on the both acid-treated CNT supports (np-CNT and wp-CNT samples) under the same operating conditions as the metal loaded samples. The main product formed at a very low conversion (1%) was methane with almost no higher hydrocarbons.

Fig. 6 illustrates the change in the activity (% CO conversion) of the Fe/np-CNT and Fe/wp-CNT catalysts over 120 h. Both catalysts reached their highest activity within 10 h. Then, they showed different activity patterns. Fig. 6 shows that Fe/np-CNT is a more active and stable catalyst compared to Fe/wp-CNT. In the case of Fe/np-CNT, syngas conversion decreased slightly from 31% to 26% in a period of 120 h, whereas, Fe/wp-CNT catalyst showed a quick drop in CO conversion from 18% to 9%.

The activity, % (CO + H₂) conversion, and product selectivity (CH₄, C₂–C₅, C₅ + and CO₂) of the catalysts after time on stream of 120 h is given in Table 4. According to the data in Table 4, the Fe/np-CNT catalyst showed a CO conversion of 26%, whereas, activity of the Fe/wp-CNT catalyst was less than 10%. In addition, C₅ + selectivity of the Fe/np-CNT (48.5 wt%) was considerably higher than that for Fe/wp-CNT (12.0 wt%) catalyst.

This study shows that the deposition of iron particles inside the nanotube pores with narrower pores within the range of studied pore diameters (higher than 10 nm) enhances FT activity and prod-

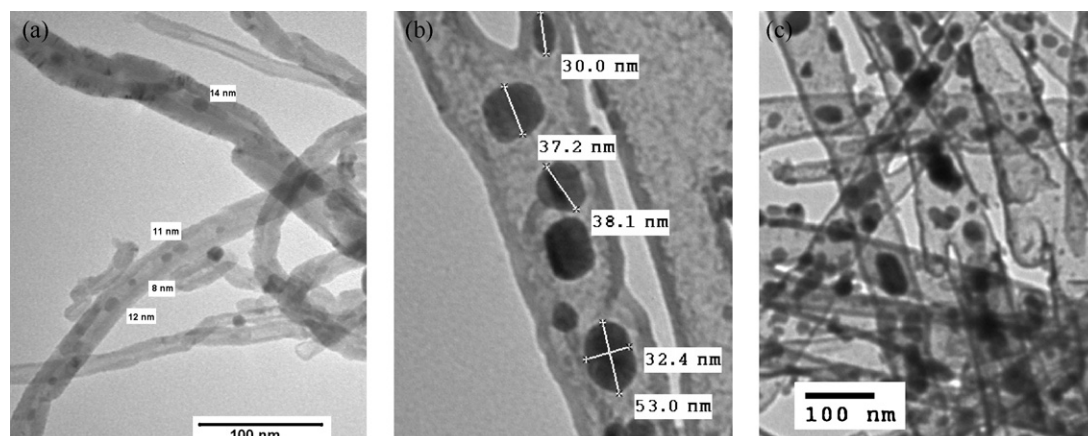


Fig. 7. TEM images of (a) used Fe/np-CNT, (b) used Fe/wp-CNT with higher magnification (c) used Fe/wp-CNT with low magnification.

uct selectivity. Several possible reasons can be associated with the improvement in the catalyst activity and product selectivity. As discussed earlier, H_2 -TPR analysis revealed that the reducibility of the Fe/np-CNT catalyst was considerably better in comparison to the Fe/wp-CNT. This phenomenon can result in the formation of more catalytically active carbide species during FTS. Besides, confinement of the reaction intermediates inside the pores can enhance their contact with iron catalysts, favoring the growth of longer chain hydrocarbons.

As given in Table 4, CO_2 selectivity for Fe/wp-CNT catalyst was less than that for Fe/np-CNT catalyst. This can be resulted from lower CO conversion for the Fe/wp-CNT catalyst. At a low CO conversion, the amount of the produced water was low which provided a lower water concentration for water–gas-shift reaction during the FT reactions.

Transmission electron microscopy analyses were conducted to study the particle size and structure of the used catalysts after 120 h of FT reactions at 275 °C. Fig. 7a and b shows TEM images of the used Fe/np-CNT and Fe/wp-CNT catalysts. As can be seen in Fig. 7a, the iron oxide particles inside the channels do not experience considerable particle agglomeration. According to several TEM of used Fe/np-CNT catalyst, the particle size distribution varied from 7 to 16 nm which is comparable to that of the fresh Fe/np-CNT (6–10 nm) catalyst. This phenomenon can be related to the interaction of the catalytic sites with the inner surface of the pores and perhaps to the spatial restriction of the CNT channels. However, as can be seen in the representative TEM image of Fe/wp-CNT catalyst (Fig. 7b and c), most of the iron species noticeably agglomerated. According to several TEM images, the most abundant particles for fresh Fe/wp-CNT (14–25 nm size range) shifted to 22–38 nm size range after the catalyst used for FT reactions for 120 h. This phenomenon resulted in decrease in FT activity of the catalyst within the reaction period and reaction conditions used in this study.

5. Conclusions

The structure and pore diameter of the CNT supports have significant influence on the catalytic performance of the catalysts in

FT process. Deposition of the metal sites inside the nanotube with narrow pore structure resulted in smaller metal particle sizes and better dispersion due to the physical restriction imposed by the CNT pores. In addition, the iron catalyst supported on the narrow pore support showed higher extent of reduction compared to its counterpart with wide pore structure. This difference is most likely due to difference in electronic properties of the inner surface of the CNTs with different diameters. Higher metal dispersion and better extent of reduction resulted in more active and selective catalysts. In the case of catalysts with metal particles inside the narrow pores, confinement of reaction intermediates inside the channels may also increase the contact time with active metal sites, resulting in production of heavier hydrocarbons.

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